

Image Processing I

Topics:

- Pixel Processing
- LSI and Convolution
- Linear Image Filters
- Non-Linear Image Filters
- Template Matching

Define img as function $f(x,y)$

Pixel Processing

$$g(x,y) = T(f(x,y))$$

ex, darken

lighten

invert

$$f - 128$$

$$f + 128$$

$$256 - f$$

low contrast

high contrast

gray

$$f/2$$

$$f \cdot 2$$

$$0.3f_R + 0.6f_G + 0.1f_B$$

Linear Shift Invariant Systems (LSIS)

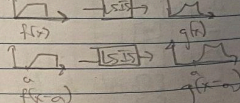
$$f(x) \xrightarrow{\text{LSIS}} g(x)$$

- linearity (like in Linear Algebra)

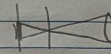
$$f_1 \xrightarrow{\text{LSIS}} g_1, f_2 \xrightarrow{\text{LSIS}} g_2$$

$$af_1 + bf_2 \xrightarrow{\text{LSIS}} ag_1 + bg_2$$

- shift invariant (output shifts w/ input)



Ideal Lens is an LSIS



Defocused img (g) is processed version of focused image (f)

- brightness variation linearity
- shift invariance (scene movement)

Convolution

- convolution of 2 functions $f(x)$ and $h(x)$:

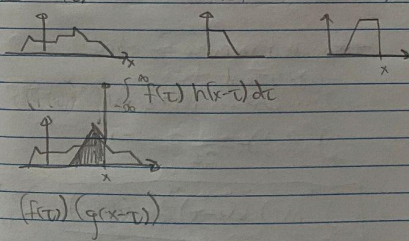
"f convolved w/ h"

$$g(x) = f(x) * h(x) = \int_{-\infty}^{\infty} f(\tau) h(x-\tau) d\tau$$

ex $f(\tau)$

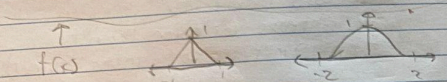
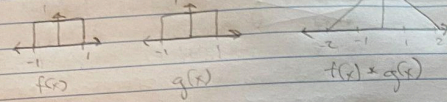
$h(\tau)$

$h(x-\tau)$



LSIS implies convolution and convolution implies LSIS

ex Convolution



$$\text{Given } f \xrightarrow{h} g \quad g(x) = \int_{-\infty}^{\infty} f(\tau) h(x-\tau) d\tau$$

Can we find h ?

What input f will produce output $g=h$

$$h(x) = \int_{-\infty}^{\infty} f(\tau) h(x-\tau) d\tau$$

→ use unit impulse function

$$\delta(x) = \begin{cases} \frac{1}{2\epsilon} & |x| \leq \epsilon \\ 0 & |x| > \epsilon \end{cases} \text{ as } \epsilon \rightarrow 0$$

$$\int_{-\infty}^{\infty} \delta(\tau) d\tau = \frac{1}{2\epsilon} \cdot 2\epsilon = 1$$

$$\int_{-\infty}^{\infty} \delta(\tau) b(x-\tau) d\tau = b(x) \quad \text{sifting property}$$

$$f \xrightarrow{h} g$$

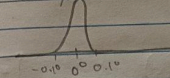
$$g = f * h$$

$$\delta \xrightarrow{h} h$$

$$h = \delta * h$$

Impulse Response of Human Eye

Distant star $\delta(x,y)$ → human eye → Point Spread Function (PSF) $h(x,y)$



human eye psf

Properties of convolution

$$a * b = b * a \quad (\text{commutative})$$

$$(a * b) * c = a * (b * c) \quad (\text{associative})$$

$$f \xrightarrow{h_1} g_1 \xrightarrow{h_2} g_2 \quad (\text{cascaded systems})$$

$$\equiv f \xrightarrow{h_1 * h_2} g_2 \equiv f \xrightarrow{h_1 * h_2} g_2$$

2D Convolution

$$f(x,y) \xrightarrow{h(x,y)} g(x,y)$$

$$g(x,y) = \sum_{m=-\infty}^{\infty} \sum_{n=-\infty}^{\infty} f(m,n) h(x-m, y-n)$$

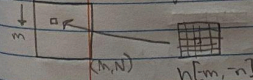
Linear Image Filters

Convolution w/ Discrete Images

$$f[i,j] \xrightarrow{h[i,j]} g[i,j]$$

$$g[i,j] = \sum_{m=-1}^1 \sum_{n=-1}^1 f[m,n] h[i-m, j-n]$$

"mask" ("kernel"/"filter")



Border Problem (3 solutions)

- ignore border (skip pixels near border)
- pad w/ const. value (e.g. avg. brightness)
- pad w/ reflection

Options

- Identity: $\delta(x,y)$
- Shift: $\delta(x-u, y-v)$
- Box Filter: $a(x,y)$

normalize vals

- images smoothed w/ box filter doesn't look "natural" (it's blocky)
- Fuzzy Filter (blur)
- use Gaussian Kernel

$$n_0[i,j] = \frac{1}{2\pi\sigma^2} e^{-\frac{1}{2\sigma^2} \left(\frac{i^2+j^2}{2} \right)}$$

σ^2 = variance
Set K (h/w of kernel) to $2\pi\sigma$
larger $\sigma \rightarrow$ more blurring

Gaussian Smoothing is separable

$$g[i,j] = \frac{1}{2\pi\sigma^2} \sum_{m=-1}^1 \sum_{n=-1}^1 e^{-\frac{1}{2\sigma^2} \left(\frac{m^2+n^2}{2} \right)} f[i-m, j-n]$$

$$= \frac{1}{2\pi\sigma^2} \sum_{m=-1}^1 e^{-\frac{1}{2\sigma^2} \frac{m^2}{2}} \sum_{n=-1}^1 e^{-\frac{1}{2\sigma^2} \frac{n^2}{2}} f[i-m, j-n]$$

Using One 2D Gaussian Filter $\equiv$ Using Two 1D Gaussian Filters

$f * \begin{bmatrix} \text{2D Gaussian} \end{bmatrix} = f * \begin{bmatrix} \text{1D Gaussian} \end{bmatrix} * \begin{bmatrix} \text{1D Gaussian} \end{bmatrix}$

K^2 multiplications vs $2K$ multiplications

Non-Linear Filters

why?

- smooth to rm img noise
- ex w/ salt+pepper noise, gaussian smears noise + removes details of what we care abt

Median Filtering

1. Sort K^2 values in windows centered at pixel
2. Assign middle val (median)

- fixes salt+pepper noise w/ $K=3$ but still loses some detail
- can't be done w/ convolutions
- but noise at every pixel requires large K that significantly reduces detail

When Gaussian is used everywhere, it blurs across edges

- we want to modify kernel based on local info
- only include pixels that are similar brightness
- "Bias" Gaussian Kernel gives lower weight to pixels w/ much diff. intensity to center

Use Bilateral Filter

- start w/ gaussian (see next pg)

$$g[i,j] = \frac{1}{W_{sb}} \sum_m \sum_n f[m,n] n_{\sigma_s}[i-m, j-n] n_{\sigma_b}(f[m,n] - f[i,j])$$

$\rightarrow$ spatial gaussian (normal blur)

$$g[i,j] = \frac{1}{W_{sb}} \sum_m \sum_n f[m,n] n_{\sigma_s}[i-m, j-n] n_{\sigma_b}(f[m,n] - f[i,j])$$

$\rightarrow$ brightness gaussian

Preserves edges! But not convolution

What is W_{sb} ?

$$W_{sb} = \sum_m \sum_n n_{\sigma_s}[i-m, j-n] n_{\sigma_b}(f[m,n] - f[i,j])$$

Increase σ_s , looks more like watercolor
Increase σ_b , becomes more like normal gaussian